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System Identification of a Small UAS in Support of Handling Qualities Evaluations

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As small unmanned aircraft systems (sUAS) have moved from the domain of the hobbyist to the defense and commercial sectors, new procedures are required to quantify vehicle performance for a designated mission. This includes consideration of the verification, validation, and certification methods that will enable a safe introduction of the UAS into the National Airspace System (NAS). A significant portion of any certification effort in the traditional manned vehicle arena includes the evaluation of handling qualities, i.e., the stability and control traits that define the ability of a pilot to safely and effectively conduct the designated mission. In the context of sUAS, aircraft may be controlled by a remote human pilot, a semi-autonomous system with remote human oversight, or a fully autonomous system. Assessing the handling qualities of a vehicle, manned or otherwise, requires both an analytical evaluation process that predicts mission performance and a flight test process that verifies mission performance. As part of the analytical evaluation, vehicle models are examined using metrics that predict handling qualities. Flight tests are then conducted to verify and validate the predicted handling qualities. When discrepancies exist, model updates are made, based on the collected flight test data. In support of an ongoing UAS handling qualities development effort led by Systems Technology, Inc., system identification focused flight test data were collected for a small fixed-wing UAS. Multiple test conditions and excitation inputs were considered, and the resulting test responses were compared against analytical models. These models were then updated based on the flight test data analysis results. Finally, the revised model responses were compared to short duration input responses from the flight test sorties.

I. Nomenclature

AGL	=	Above Ground Level
COA	=	Certificate of Authorization
FMU	=	Flight Management Unit
FREDA	=	FREquency Domain Analysis (STI software)
GPS	=	Global Positioning System
IMU	=	Inertial Measurement Unit
MTE	=	Mission Task Element
NAS	=	National Airspace System
OMS	=	Orthogonal multi-sine

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RC	=	Radio Controlled
STI	=	Systems Technology, Inc.
UAS	=	Unmanned Air Systems
UAV	=	Unmanned Aerial Vehicle
UMN	=	University of Minnesota
sUAS	=	small Unmanned Aerial Systems

II. Introduction

Unmanned Air Systems (UAS) are here. Operators, from first responders to commercial entities, are demanding access to the National Airspace System (NAS) for a wide variety of missions. New business models involve a proliferation of small UAS (sUAS) that will operate beyond line of sight at altitudes of 500 ft and below. A myriad of issues continues to slow the development of verification, validation, and certification methods needed for the safe introduction of UAS to the NAS. The issues include the lack of a consensus on the UAS categorization process, as well as the lack of quantitative certification requirements, including the definition of handling qualities. The "how to" of safely integrating UAS in the NAS raises many questions. To date, there have been few answers, due to problem complexity. However, the lack of quantitative data has been one important contributing cause of the failure to achieve consensus. The interested parties now include traditional airframers, established UAS manufacturers, academic institutions, and many newcomers, such as Amazon, Google, and Facebook that see UAS as a means to various commercial ends. The program undertaken by Systems Technology, Inc. (STI) does not propose to tame the entire verification, validation, and certification problem, but instead aims to address the important need to define UAS handling qualities in both piloted and autonomous operations.

Fig. 1 illustrates the proposed process that begins with UAS classification. Because of the wide variety of UAS types (fixed wing, rotary wing on traditional helicopters, multi-rotor configurations, ducted fans, airships, etc.) and vehicle sizes (from micro vehicles to the Global Hawk with a wing span of a Boeing 737), a one-size-fits-all set of requirements is infeasible. In the proposed process, given an appropriate classification, a mission in the form of mission task elements (MTEs) is next considered (Ref. [1]). Mission examples include air-to-air and/or air-to-ground sensor tracking for surveillance, terrain surveying for pipelines or agriculture, entertainment or real estate filming, weather monitoring, package delivery, and wifi access. Missions are then broken down into specific task elements that are common to all missions (e.g., takeoff and landing), as well mission-specific elements (e.g., precision target tracking, high altitude loiter, obstacle avoidance, etc.) These mission task elements are used to identify specific criteria that predict handling qualities analytically and test demonstration maneuvers that verify handling qualities in flight. Further details of this process can be found in Ref. [2].

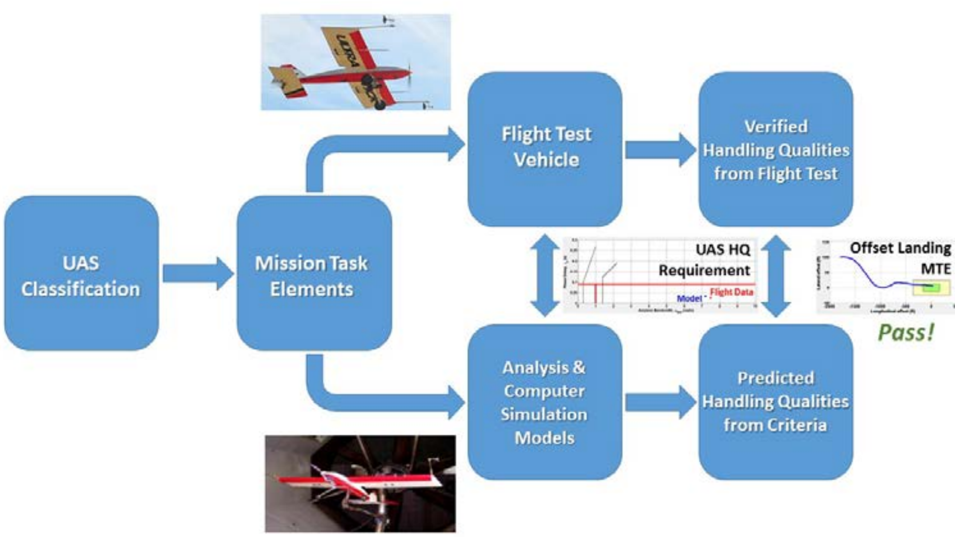


Fig. 1: The Proposed UAS Handling Qualities (UAS-HQ) process.

This paper documents the system identification fixed wing flight test campaign flown by the University of Minnesota (UMN) using an UltraStick 120 research UAS. While the test techniques, system identification, and model update processes are standard to the piloted vehicle industry, their application to UAS is relatively new and largely untested as part of a MTE based UAS handling qualities assessment process. The flights address the “Flight Test Vehicle” block in Fig. 1, where automated system inputs are used to collect flight test data. The collected data were then used to identify the flight vehicle and update the actuator and vehicle models. The process is detailed below. Also included here is a review of the data collected, the input type and the performed identification. The analysis illustrates how different input types and their identification compare against the updated models. While the test flights addressed pitch, roll and yaw, to simplify exposition, only the pitch axis will be examined. The results and conclusions determined for pitch are applicable to roll and yaw. The understanding gained from these evaluations will be used to guide future sUAS test flights and vehicle evaluations.

III. Flight Test Campaign

The flight test campaign in support of the identification and model update effort was flown at the University of Minnesota between the Fall of 2017 and Spring of 2018.

A. Objectives

The objectives for the entire projects flight test program were as follows:

- Use several input excitation command types (frequency sweeps, multisines, and short duration inputs) to identify the vehicle and actuators.
  - The database will be used to determine the effectiveness of each command input, leading to recommendations that will guide future sUAS tests.
- Demonstrate an efficient process for determining aircraft flying and handling qualities metrics for UAS via flight testing.
- Generate a flight test database using a variety of flight test inputs for a fixed-wing sUAS at multiple flight conditions and aircraft configurations.
  - Collected data will be sufficient to determine aircraft flying and handling qualities metrics at each flight condition and configuration.
  - The database will include a nominal baseline configuration and two off-nominal configurations (e.g., a configuration with added time delay and a configuration with an unfavorable c.g. shift).
- Use the resulting flight test database to compute flight-derived handling qualities parameters and compare them with those obtained analytically.

The primary focus in the effort reported in this paper was the first bullet: examination of a range of excitation commands for UAS system identification.

B. Test Facilities

The flight test facilities, shown in Fig. 2, consist of the UAV laboratory on the UMN campus and the airfield located at the UMore Park Test Range. The aircraft maintenance and pre-flight checkouts (simulation-in-the-loop, hardware-in-the-loop, etc.) occur in the UAV laboratory. The flight tests took place at the UMore Park Test Range near Rosemount, MN. The Test Range is located on sparsely populated agriculture fields owned by the UMN. Flights are conducted year-round. A winter flight is shown in Fig. 2.

The UMore Park Test Range is located within Class G airspace. The test aircraft must remain within the area centered at 44º 43’32.71”N and 93º 4’44.49”W, with a radius of 0.28 NM and a ceiling of 400 feet Above Ground Level (AGL). All flights are coordinated with the Rosemount Research and Outreach Center Manager. In addition, the University of Minnesota UAV lab has obtained the proper Certificates of Authorization (COAs) for legal operation of UAS flights at the UMore Test Range. Non-participants are kept clear of this zone during operations. The dimensions comply with the 1-mile lateral and 400 foot ceiling limits and were created to avoid structures and major roadways while staying within direct line of sight of the operations crew. Restriction to the flight research zone minimizes risk to non-participants. The Operations Lead, Pilot, and Observer ensure that the aircraft remains within the flight research zone and will abort the mission if there is a risk of exiting the area. Failsafes on the aircraft ensure that the aircraft will remain in the test area in the event of link loss. Launch and recovery are conducted from a 900 foot by 60 foot turf runway aligned in an East-West direction. A right pattern is used when launching/recovering in an east-bound direction and a left pattern is used when launching/recovering in a west-bound direction. These patterns are used to avoid overflying the operations crew.

a) Laboratory

b) Airfield

Fig. 2: UAS flight test facilities at UMN.

C. Test Articles

The UAS laboratory at UMN maintains the UltraStick series of fixed wing UAS to serve as testbeds in several ongoing research projects such as control law design, navigation and guidance and fault detection, isolation and reconfiguration (Ref. [3]). The series consists of the UltraStick 120, UltraStick 25e and the UltraStick mini. The laboratory also supports UAS related research of other organizations by providing flight-testing services on these testbeds. Two vehicles were flown as a part of this test campaign, the UltraStick 25e and the UltraStick 120 and are shown in Fig. 3. The flight data shown and discussed here are from the UltraStick 120 only. While only the UltraStick120 results are reported on here, this vehicles design is widely used in the sUAS industry and is representative of a broad category of vehicles.

1. UltraStick 120

The UltraStick 120 was a commercially available, RC hobbyist airframe. The UMN UAS Lab converts these airframes into UAS platforms for research. Relevant size and performance values for the UltraStick 120 are included in Table 1.

The aircraft is equipped with a commercial RC hobbyist control system (propulsion system, batteries, receiver, and servos) for manual control as well as additional systems for data collection and autonomous flight modes. A system-level design has been implemented that allows the pilot to take manual control of the aircraft. Further, the manual control system failsafe defaults to a descending, power-off spiral to keep the aircraft contained within the test range.

Fig. 3: UltraStick 120 (center) and UltraStick 25e aircraft.

It is powered by an electric motor and has removable wings for transport. The UltraStick 120 has a conventional horizontal and vertical tail with rudder and elevator control surfaces. The aircraft has a symmetric airfoil wing with aileron and flap control surfaces. The rudder and elevator are actuated by Hitec HS5245MG servos. The flaps and ailerons use Hitec HS5625MG servos. The aircraft is propelled by a 1900W Actro 40-4 brushless electric motor with a Graupner 14 x 9.5 folding propeller. Power for the motor comes from two 5000mAh 5-cell lithium polymer batteries connected in series. The servos are powered by a separate 1350 mAh 3-cell lithium polymer battery. The main internal payload bay is located in the fuselage, directly under the wing; additional payloads may be accommodated in the aft fuselage or externally. The UltraStick 120 aircraft has space aft of the avionics bay for additional sensors or payloads. sUAS with these general specifications and configuration are typical of the sUAS community at large

Table 1: UltraStick 120 Aircraft Parameters

Wingspan	1.92 m
Chord	0.43 m
Length	1.73 m
Airframe Weight	6.0 kg
Max Weight	10.0 kg
Cruise speed	23 m/s (typical)
Stall speed at Max Weight	13 m/s
Airspeed range	10-41 m/s

2. Goldy3 Flight Control System

The Goldy3 flight control system was designed to integrate with a RC hobbyist control system and allow complete autopilot functionality. The system was developed in house by the UMN UAV Lab. The system allows for data collection and autonomous flight modes. A system-level design has been implemented that allows the pilot to take manual control of the aircraft via a programmed micro-controller. A functional diagram of this system is in Fig. 4.

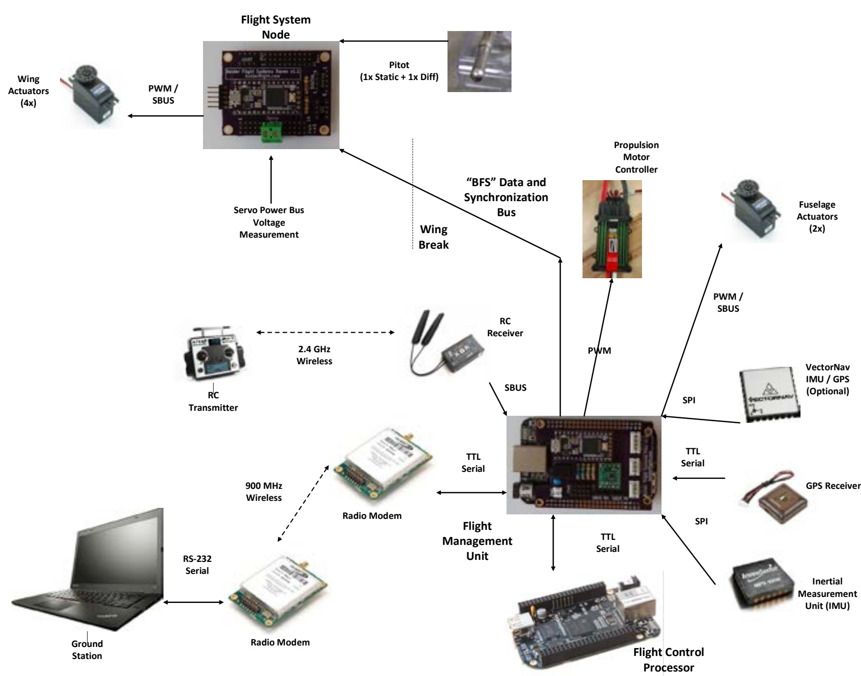


Fig. 4: Avionics functional diagram, based on UltraStick 25e installation.

Commands are sent to the aircraft via a hobbyist RC transmitter and receiver. The receiver sends commands and communicates with the FMU via a SBUS-2 protocol. The FMU controls the "safety"/"auto" switch; in "safety" mode, commands from the receiver are forwarded to the servo actuators. In "auto" mode, commands from the flight computer are sent to the servo actuators. The flight computer receives flight data from sensors including GPS, Inertial Measurement Unit (IMU), and pressure transducers. Data from the flight computer is downlinked through a radio modem to a ground control station where researchers can monitor its operation.

At any time, the pilot can revert to the “safety” mode via the “safety/auto” switch. Additionally, failsafe commands are set in the RC receiver and FMU such that in the event of a lost link, the receiver switches the aircraft to “safety” mode, cuts power to the motor, and puts the aircraft in the previously mentioned descending spiral.

3. System Architecture

For system identification analysis, the system architecture shown in Fig. 5 was used. The bare-airframe (or plant) is noted by A/C, the actuators are identified by ACT, and any flight control system or feedback compensators are defined by C and FB respectively. Each of A/C, ACT, C, and FB are in general multi-input-multi-output systems. The pilot commands are defined by the vector r and denote commanded aircraft states (e.g., pitch-command or roll-command). In some cases, the computer may command some of the r inputs (e.g., for velocity or speed-command inputs). The commands to the actuators are defined by the vector u. The vector δ defines the bare-airframe control surface deflections. Both u and δ also include the throttle command and input. The u_{ex} signal is the location where computer-commanded input excitations are applied.

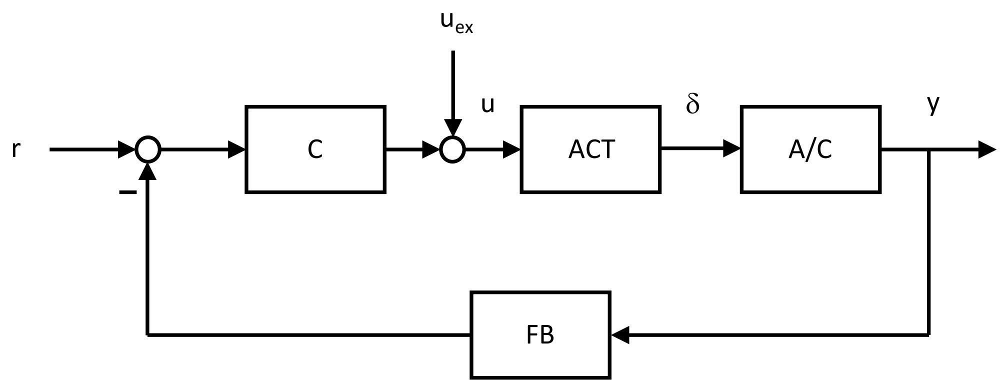


Fig. 5: System architecture.

4. Vehicle Simulink Models

The UMN laboratory also provided MATLAB/Simulink-based (Ref. [4]) simulation models for the UltraStick series. These models are typically used to obtain linear state-space models at different trim conditions, to enable control/navigation and other function software design, and to carry out simulations in closed loop with the developed software before actual flight-testing. Depending on the UltraStick UAS simulated, these models have nonlinear rigid body dynamics coupled with either a linear, coefficient-based aerodynamic model or a look-up-table-based nonlinear aerodynamic model. Data for the look-up table have been obtained from a wind tunnel test of the UltraStick mini and other additional and extensive testing of the UltraStick 120.

The Matlab/Simulink UAS simulation package provided by UMN is capable of modeling 3 different airframes: 1) UltraStick 120, 2) UltraStick 25e and 3) miniMUTT (flex wing UAS). The simulation package consists of different Simulink models and accompanying Matlab scripts, .mat files and Simulink Library blocks. The Library blocks are key to maintaining consistency across the different models and making it easier to modify or swap different blocks.

The UltraStick 120 bare airframe models feature standard aircraft inputs, states, and outputs. The inputs include throttle, elevator, rudder, left/right aileron, and left/right flaps. The states include the attitudes, rates, velocities of the vehicle, and Earth relative position. The outputs include the body attitudes, rates, accelerations, airspeed, angle of attack, flight path angle, sideslip, and altitude.

A simple first-order actuator is modeled for the elevator, aileron, rudder, and flap surfaces. The throttle is non-dimensional and limited to 0-1. The actuators for each control surface were found using the methods described in Section IV and are given in Table 2.

The aircraft controller implements a simple pitch and bank attitude tracker. The pitch attitude tracker is shown in Fig. 6. The internals of the Pitch Tracker block are shown in Fig. 7. The Roll Tracker block is an exact duplicate, but with unique gains. A yaw damper was included as a simple washout filter in the yaw rate to rudder feedback loop. As flown with the UltraStick 120, the pitch axis featured an attitude command system with speed hold, while the roll axis featured an attitude command and the yaw axis – a rate command system.

Table 2: Actuator Models

Elevator	Aileron	Rudder
$\frac{\delta_e}{\delta_{ec}} = \frac{8}{s+10} e^{-0.065s}$	$\frac{\delta_a}{\delta_{ac}} = \frac{40}{s+12} e^{-0.060s}$	$\frac{\delta_r}{\delta_{rc}} = \frac{10}{s+10} e^{-0.065s}$

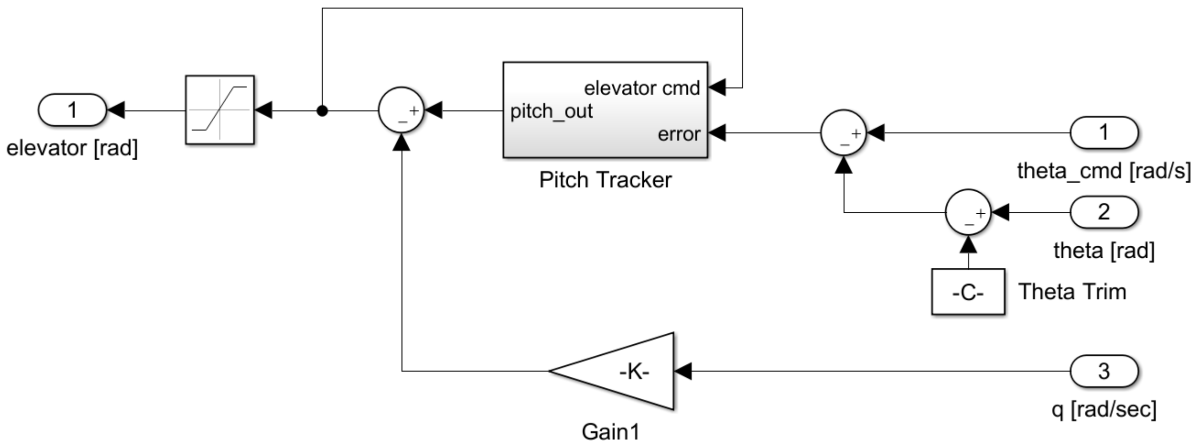


Fig. 6: Pitch attitude tracker.

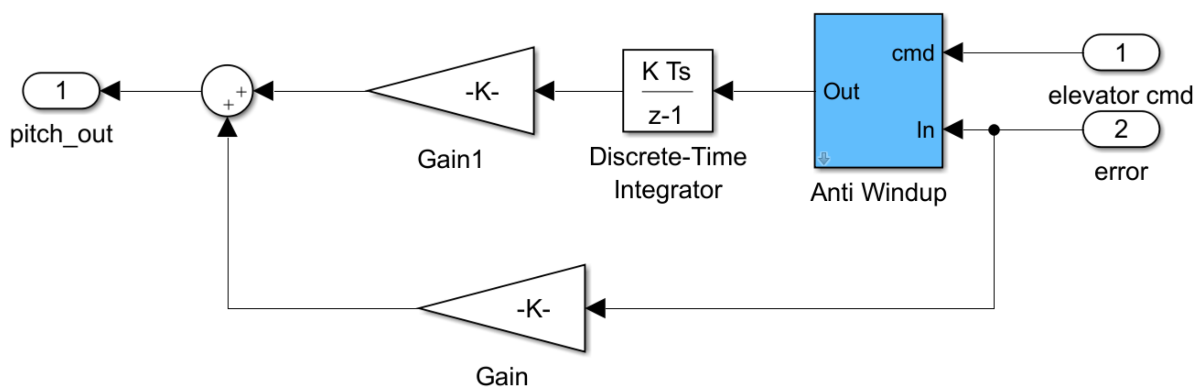


Fig. 7: Representative tracker block.

D. Test Description

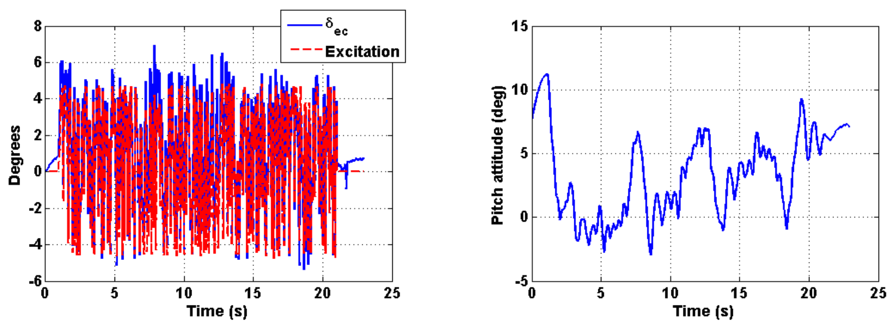
A nominal baseline configuration and two additional off-nominal configurations were flown. All configurations included the same flight control system feedback augmentation response types as identified above. The off-nominal configurations were: 1) added delay and 2) a configuration with an unfavorable c.g. shift. Two airspeed flight conditions were flown. The primary cruise airspeed for all evaluation tasks was 23 m/s and the primary approach airspeed for all evaluation tasks was 17 m/s. The primary altitude for all cruise and approach evaluation tasks was approximately 50 m. The defined "approach" condition is not a true approach condition because it was flown at a constant altitude, but at a lower airspeed, with 50% flap deflections to reflect a true approach condition. Maintaining a constant altitude for the defined approach condition was done to simplify and minimize risk for system identification flight test points and to preserve the desired test conditions. No specific flight patterns were required, other than to remain in visual line of sight. The vehicle was flown in circuits around the test area with 700 m straight legs and 45° turns at the constant 50 m altitude. All evaluated test conditions were flown by the on-board flight computer, emulating an autonomous or semi-autonomous system. The test inputs were preprogrammed into the flight computer and initiated by the remote UAS operator. The operator limited his inputs to small, low frequency commands that were intended to maintain the test condition.

E. Excitation Signals

This section provides descriptions of each of the input profiles and includes sample time histories of both the excitation command, the shaped command that was sent to the actuator, and the attitude responses. The profiles were uniform across each axis. Here, pitch axis examples are shown. These examples represent the computer-generated inputs and the resulting vehicle responses. Each of the sample commands includes both a plot of the original excitation command u_{ex} (red dashed line), and the resultant shaped surface command u (solid blue line).

1. Multi-Sine

The orthogonal multi-sine (OMS) (Ref. [5]) input profiles are mutually orthogonal in the time and frequency domains and completely uncorrelated. These OMS were applied to each axis independently, the elevator and aileron in combination, and all three axes in combination, elevator, aileron and rudder. Each OMS was 20 seconds long, had a 4 deg amplitude and covered a frequency range from 1-50 rad/s. Samples of the pitch command and attitude response are provided in Fig. 8.

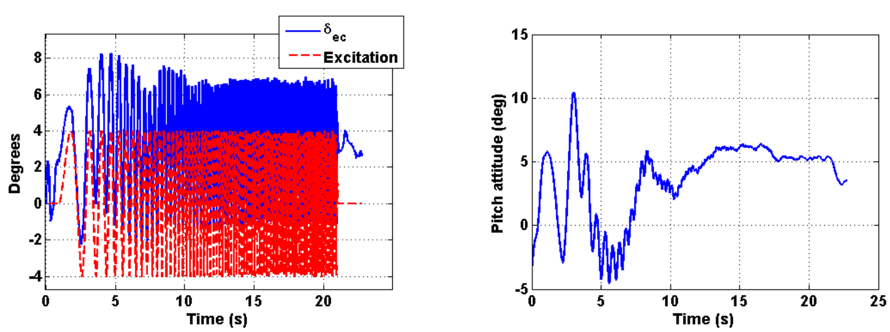


a) OMS Command Time History b) Pitch Attitude Time History

Fig. 8: Example orthogonal multi-sine input and resulting aircraft output.

2. Linear Chirp

The linear chirp input excitation was designed to be 20 seconds in duration, have an amplitude of 4 deg, and cover a frequency range of 1 to 50 rad/s. This was applied to each axis independently, and the aileron and elevator in combination. For the input to the elevator and aileron in combination, the pitch excitation had increasing frequencies over time, while the roll excitation started at high frequency and decreased to the lower frequency limit over time. This provided separation in the frequencies between the axes. Samples of the pitch command and attitude response are given in Fig. 9.

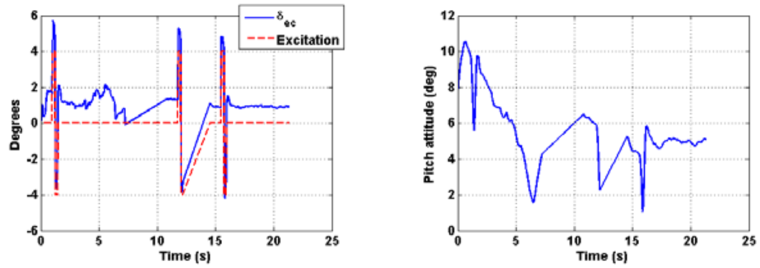


a) Chirp Command Time History b) Pitch Attitude Time History

Fig. 9: Example chirp input and resulting aircraft output.

3. Doublet

The doublet excitation profile was designed to have a 4 deg amplitude. The pulse width varied based upon the axis of the input. For every axis, the pulse width (half of the total doublet width) was designed to target a particular frequency through the following relation: 0.7/(2*target frequency). The target frequency is in Hz. For pitch, the target frequency was the short period mode, estimated at 1.51 Hz. For the aileron and rudder doublet, the Dutch roll frequency was targeted and estimated to be 0.65 Hz, Samples of the pitch command and attitude response are given in Fig. 10.

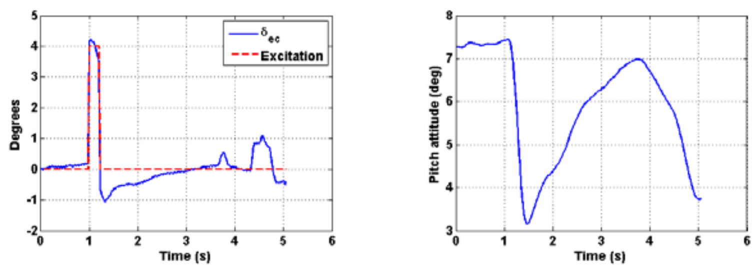


a) Doublet Command Time History   b) Doublet Pitch Attitude Time History

Fig. 10: Example doublet input and resulting aircraft output.

4. Pulse

The pulse input was designed to have a 4 deg amplitude. The pulse width was defined in the same manner as the doublet. Samples of the pitch command and attitude response are given in Fig. 11.

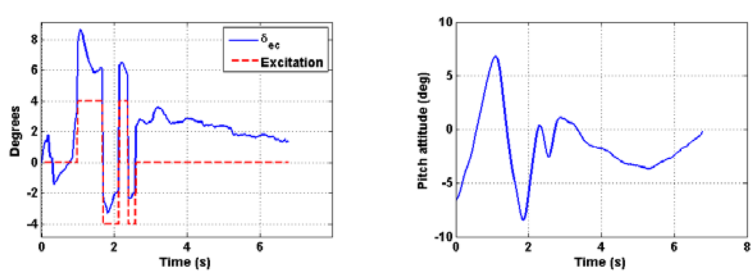


a) Pulse Command Time History   b) Pulse Pitch Attitude Time History

Fig. 11: Example pulse input and resulting aircraft output.

5. 3-2-1-1

The 3-2-1-1 input is a set of pulses of varied widths. A base width, the "1" in 3-2-1-1, is defined in the same manner as the pulse and doublet widths, based on the short period and Dutch roll frequencies for the pitch and roll axis, respectively. The "2" and "3" are then double and triple the base width pulses. For example, if the base width was 1 second, the first pulse would be 3 seconds wide, the next 2 seconds wide, followed by two pulses of 1 second each. Each new pulse reverses direction from the prior one. Samples of the pitch command and attitude response are given in Fig. 12.



a) 3-2-1-1 Command Time History   b) 3-2-1-1 Pitch Attitude Time History

Fig. 12: Example 3-2-1-1 input and resulting aircraft output.

IV. Model Update Process

The model update process is a key component of any handling qualities assessment program, manned or otherwise. As accurate as models can be, they are only representations of the true flight vehicle. The process defined below was conducted to both validate and update, if required, the existing analytical models, based upon the newly collected data.

The identification of each set of data was performed using STI's FREquency Domain Analysis (FREDA) software (Ref. [6]).

A. Bare Airframe

The process for updating the bare airframe dynamics of the vehicle is as follows:

- Compare the frequency response identified from the flight test data with the original UltraStick 120 model.
- The bare airframe dynamics were identified from the flight test data using the following input/output time histories:
  - Pitch: Input – Measured elevator actuator position; Output – Measured pitch rate
  - Roll: Input – Measured aileron actuator position; Output – Measured roll rate
  - Yaw: Input – Measured rudder actuator position; Output – Measured yaw rate
- Identify any model discrepancies relative to the flight test data and adjust the modal parameters in each axis as appropriate to achieve an improved match to the flight test data. Adjust the model gain as required. The relevant modal parameters are discussed below.

For the longitudinal axis, the phugoid and short period frequency and damping ratio were considered for adjustment though most conditions only required adjustments to the short period. For the lateral-directional axis, the roll mode and the Dutch roll frequency and damping were considered for adjustment. Most lateral-directional cases only required adjustments to the roll mode.

B. Actuator Models

A similar process was used to modify and update the actuator dynamic models and is given as follows:

- Compare the frequency response identified from the flight test data of the actuator (actuator command to actuator position) with the frequency response of the actuator model, a 1st order transfer function with added delay. The magnitude response was matched first by modifying the inverse time constant of the actuator model until a suitable overlay of the identified data was found. This also included adjustments to the actuator gain as well.
- The phase response was then modified by adding delay to the transfer function in the form of e^{-st} where t is the added delay.
- With the phase response matched, the flight data and the actuator model were compared once again to ensure a desired fit was found.

C. Longitudinal Model Update

This section describes the changes made to the bare airframe and actuator models for the longitudinal axes. As stated previously, the steps are the same for the lateral/directional axis, although with different terms to adjust. The flight test data in this section are from a baseline flight test conducted in straight and level flight, 23 m/s airspeed and 100m altitude. The UltraStick120 model used in the comparison was trimmed at these flight conditions.

The changes in the bare airframe were made to the pitch rate in elevator (q/δ_e) transfer function. A comparison of the original plant model, new plant model, and the flight test frequency response generated from a chirp input is shown in Fig. 13. The original and updated longitudinal aircraft dynamics are provided in Table 3. Note that the gain of new model represents an increase of 1.5 times the original model value. Only the short period dynamics were altered, increasing the frequency slightly while reducing the damping. The adjustments to the elevator actuator model are shown in Fig. 14. The updated transfer function is shown in Table 4. The response of the longitudinal system, including aircraft and actuator dynamics is shown in Fig. 15.

Table 3: Original and New Values of Longitudinal Plant Dynamics

	Original	New
ω_{sp}	9.48 (rad/s)	12.5 (rad/s)
ζ_{sp}	0.753	0.5
Gain	-	1.5

Table 4: Original and New Elevator Actuator Model

Original	New
$\frac{\delta_e}{\delta_{ec}} = \frac{50.27}{s+50.27}e^{-0.05s}$	$\frac{\delta_e}{\delta_{ec}} = \frac{8}{s+10}e^{-0.065s}$

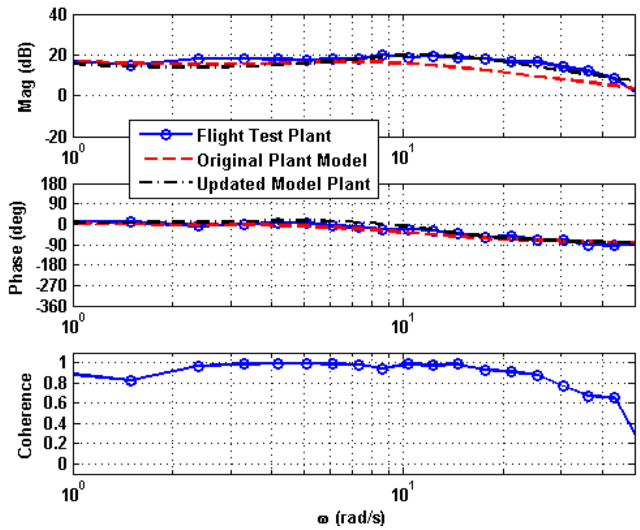


Fig. 13: Original and updated q/δₑ response compared against flight test.

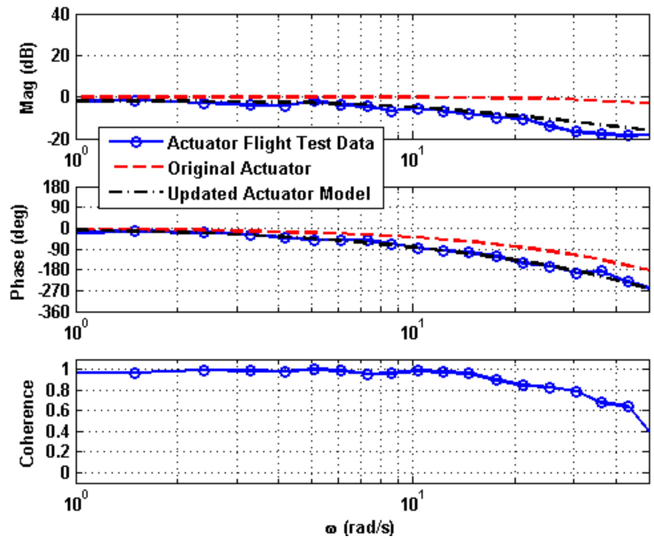


Fig. 14: Original and updated elevator actuator models compared with flight test.

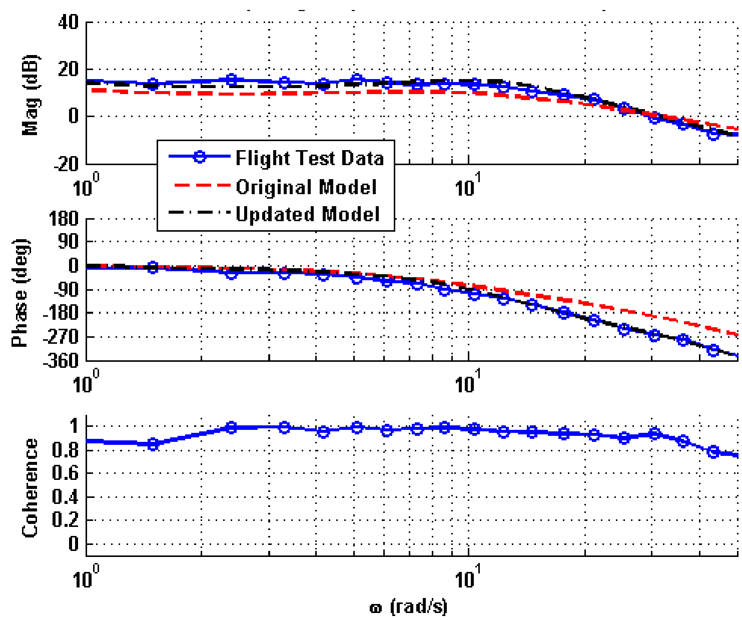


Fig. 15: Original and updated longitudinal models (actuator+aircraft) compared with flight test.

D. System Identification Comparison

In support of the flight test objectives, each input profile described in the Excitation Signals section was used to identify the flight vehicle from the test data. The excitation input was summed with the compensated feedback signal to generate the actuator input signal "u", as shown in Fig. 5. The output pitch rate was used to identify the q/δ_{ec} response. This identification returns the actuator and bare airframe dynamics of the vehicle and is not a complete closed-loop identification. A subset of the flight test card is provided in Table 5, highlighting only the pitch axis evaluations.

The overlay of each of these identifications, including the Simulink model frequency response, is shown in Fig. 16. The displayed model response reflects the updated model response discussed previously. The updated elevator and q/δ_e dynamics are given below in Table 6.

Table 5: Flight Test Card

Leg	Input Name	Description	Duration (sec)	Amplitude (deg)	Start-End Frequency (rad/s)
1	OMS_ele1	Orthogonal multi-sine applied to elevator	20	4	1-50
4	OMS_elevail1	Orthogonal multi-sine applied to elevator and ailerons in combination	20	4	1-50
5	OMS_3axes1	Orthogonal multi-sine applied to elevator, ailerons and rudder in combination	20	4	1-50
6	chirp_elev1	Linear chirp to the elevator	20	4	1-50
9	chirp_elevail1	Simultaneous linear chirp to the elevator and ailerons (ele increasing freq, ail decreasing)	20	4	1-50
Leg	Input Name	Description			Pulse Width (s)
10	doublet_elev	A doublet input to the elevator	20	4	0.23 (half width)
13	pulse_elev	A pulse input to the elevator	20	4	0.23
17	3211_elev	A combination of 3 width, 2 width, 1 width, 1width elevator pulses in opposing directions for each new input	20	4	0.23 (the "1" width)

Table 6: Baseline Longitudinal Aircraft and Elevator Actuator Models

Elevator Actuator Model	Longitudinal Model
$\frac{\delta_e}{\delta_{ec}} = \frac{8}{s+10} e^{-0.065s}$	$\frac{q}{\delta_e} = \frac{75.03(-0)(0.3977)(5.966)}{[0.3078, 0.5362][0.5, 12.5]}$ [1]

An encouraging result from this evaluation is the relative uniformity of the identified systems, even in the presence of excitations of the other control surfaces. This result suggests that the OMS input excitation can be performed in multiple axes without interfering with or contaminating the results of the others, as has been exemplified by Morelli (Ref. [5]) and others. This characteristic means that fewer numbers of flights are required to perform the identification of the vehicle, saving time and cost during the test campaign. As for the OMS identification, the linear chirp exhibits the ability to have multi-axis input excitations not adversely affect the system identification. The short duration inputs also provided relatively close matches of the modeled system to the rest of the identifications. As expected, the coherence for these identifications was lower at the high frequencies and did have some variance in the lower frequencies from the OMS and chirp input signals. The doublet had the greatest mismatch of the magnitude at the lower frequencies; and the pulse and doublet had some trouble matching the magnitude and phase at the highest frequencies.

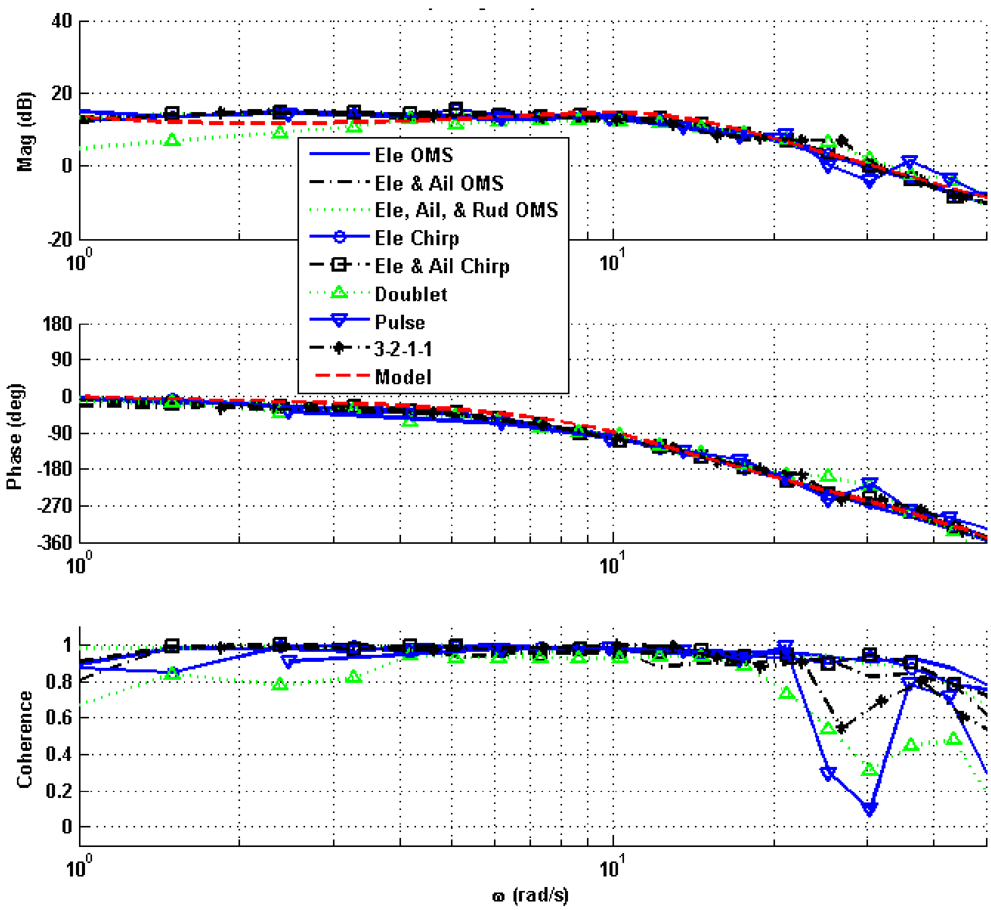


Fig. 16: q/δ_{ec} identification – all methods.

E. Model Update Validation

To validate the updated model, simulated time history responses of the model were compared against flight test responses generated from short duration inputs for the same test condition. These short duration inputs were the same

[1]. The STI shorthand form of displaying a transfer function is defined by: $a(s+b)[s^2+2\zeta\omega s+\omega^2] = a(b)[\zeta,\omega]$

doublets, pulses, and 3-2-1-1s described above. The actual flight test inputs were used to generate the model responses, so a true one-to-one comparison could be performed.

The short duration responses are shown in Fig. 17, Fig. 18, and Fig. 19. Two sets of data dropouts from ~7-11 seconds and ~12-15 seconds region affected the later portion of the doublet maneuver. Even in the presence of these dropouts, the model response for the doublet excitation exhibited a close match to the excitation flight data. The pulse and 3-2-1-1 excitation response demonstrated a good model fit as well; however there is a modest amount of amplitude mismatch. This mismatch, which occurs for the lower frequency inputs, is likely due to system nonlinearities, such as actuator free play. On the whole, these responses validate the model updates made above and the process used to generate the model revisions.

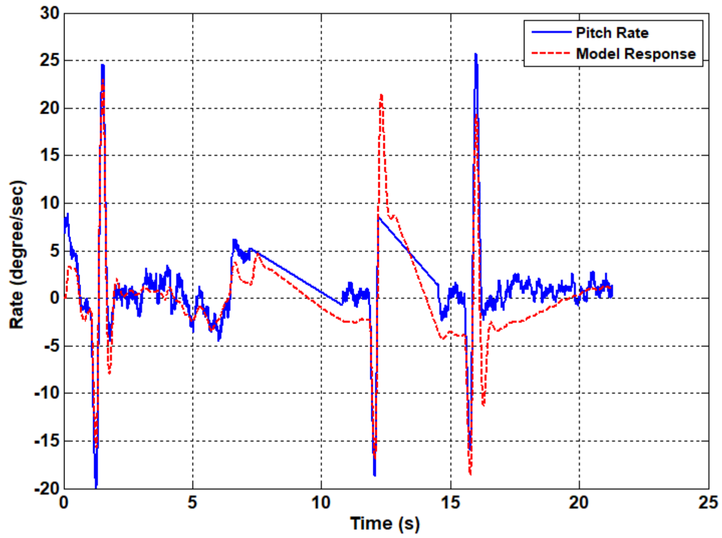


Fig. 17: Doublet response.

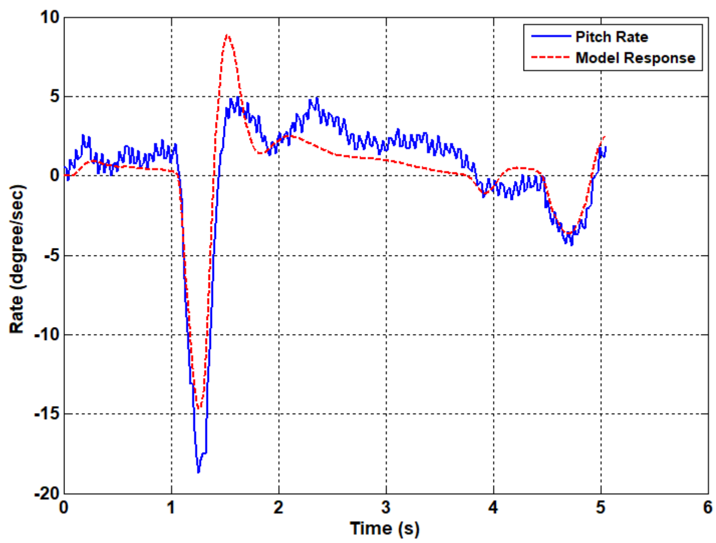


Fig. 18: Pulse response.

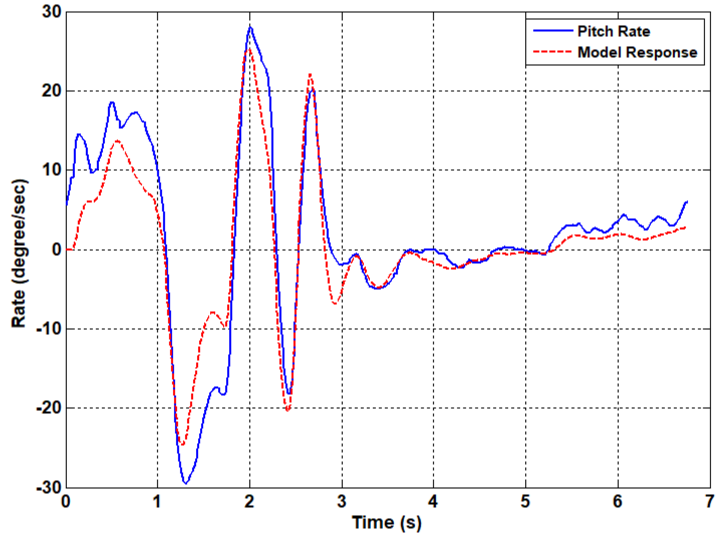


Fig. 19: 3-2-1-1 response.

V. Concluding Remarks

The Flight Test Vehicle block of the proposed UAS handling qualities process was executed on an exemplar sUAS. The system identification and model update process component were successfully demonstrated on the UltraStick120, providing an initial validation of the system identification and model update component of the proposed UAS handling qualities process. Multiple excitation input profiles were considered including chirps, multi-sine and short duration inputs. The chirp inputs were demonstrated as an effective identification method suitable for application to identifying and updating the vehicle dynamics models. The following conclusions were found based upon the analysis presented above:

- The orthogonal multi-sines and chirp inputs provided consistent system identification results.
- The orthogonal multi-axis inputs provided an effective excitation means across the axes reviewed with no undesirable artifacts.
- The short duration inputs provided close matches with the OMS and chirp input profiles in a limited frequency range around the dominant mode for a given axis (e.g., the short period), but as expected, input power degraded at the higher and lower ends of the frequency region of interest.
- The short duration inputs (doublet, pulse, 3-2-1-1) generated using the updated model, applying the test inputs from flight, closely matched the vehicle response from flight test.
- The techniques described here have long been used to identify the characteristics of manned aircraft for handling qualities assessments. The results described in this paper establish the ease of using these methods to properly characterize sUAS as part of a process that will ultimately reveal the suitability of the handling qualities of a selected vehicle for a given mission.

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